Low-cost chaotic radar design

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ABSTRACT

An approach for creating a low-cost Chaos Pulsed-Doppler Radar is presented. The objective of this effort is to develop a practical realization of a Chaotic Radar with performance advantages over other approaches. Many groups[1], [2], [3] have proposed that Chaotic Waveforms are an effective radar signal generator due to: the relatively low cost of producing complex wideband waveforms and the difficulty in detecting and spoofing inherently complex modulations. PRA and Duke University report on the development of a radar design that uses a novel high-speed chaotic waveform generator. Preliminary experimental results are presented that characterize the performance of a chaotic waveform generator. In addition, the radar architecture will be proposed, realistic radar design criterion will be set forth, and simulations of a complete radar will be used to compare the chaotic radar to more traditional radar approaches.

Keywords: Chaos radar, chaotic waveform generator, Pulse Doppler Radar, random-signal radar, low probability of detection, low probability of intercept, coherent reception, pulse compression

1. INTRODUCTION

With the erosion of available radar spectrum due to the proliferation of commercial WiFi and cellular phones, Code Division Multiplex Access (CDMA) waveforms have taken on new importance in the radar community. Chaotic waveforms can be viewed as a type of CDMA waveform that offer the potential to provide multiple radar usage in this reduced RF spectrum environment. Using chaotic waveforms for radar has been an area of active research since at least 1998 [4]. The motivation for this research has been driven by three properties of chaotic signals: "random" time evolution with no periodicity, fairly even distribution of the waveform energy over a continuous range of frequencies, and the ability to be generated from simple deterministic systems. The challenge for chaos waveforms is whether these properties are sufficient to provide unique and effective radar applications. A joint project between PRA and Duke University has shown that the three properties of chaos can be used to make a radar waveform that is simple to implement, easy to process, and less susceptible to interference or jamming (electronic attack) [5]. Specifically, PRA and Duke designed a pulsed-Doppler radar architecture that uses a digital signal from a time-delay feedback chaotic waveform generator (CWG).

While chaos-based radar has been considered for many applications such as through-the-wall radar [3] and SAR [6], a promising near-term application is in dismount detection/perimeter surveillance/situational awareness radars, where the essential properties of chaotic waveforms can be applied to enhance radar performance without the expense of extensive signal processing capabilities and at a low cost. The first property is *flexibility* in that chaotic waveforms can support both homodyne and heterodyne architectures, which allows these waveforms to be implemented in any radar design. The second property is *orthogonality*, which allows the use of multiple simultaneous chaotic waveforms due to low cross correlation of distinct chaotic waveforms. Finally, the third property is *low-probability of detection*, which results from the randomness of the waveforms that spread the signal energy out over a wide spectrum.

In this paper, preliminary results from the development of the prototype CWG are presented as well as the simulation performance of a pulsed-Doppler radar design using the CWG waveform benchmarked against a pseudo-random digital waveform and a Kasami *m*-code waveform. In a few instances, the digital waveforms from the time-delay feedback CWG perform better than random digital waveforms in resolution, jamming resistance, cross-correlation, and auto-correlation. In addition, the CWG waveform performed better in a pulsed-Doppler system than the more structured binary phase (Kasami) codes, mostly due to the inherent variability in the waveform. A summary of the results of the waveform comparisons is shown in Table 1. Three waveforms are compared: CWG, digital noise, and a Kasami code. In each case, the waveforms have the same bandwidth (null-to-null) and pulse length.

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Table 1. Summary of waveform comparison results.

	CWG	Noise	Kasami
Bandwidth (MHz)	42	42	42
Code length (N)	<255	255	255
Clocked transitions	No	Yes	Yes
Peak to RMS sidelobe (single) (dB)	20[3]	23[3]	23
Peak to RMS sidelobe (dwell) (dB)[1]	32[3]	39[3]	23
Peak to 1st sidelobe (dB) [1,2]	13[3]	39[3]	23
Cross-correlation (dB) (loss over Welch Bound)	~2	~1.5	~0.5
Waveforms in set	∞	2^N	~Sqrt(N)
Spoofing tolerant	Yes	Yes	No
Narrowband jamming tolerant	Yes	Yes	Yes

1. Averaged over 44 pulses
2. Bandwidth limited result
3. Not an ensemble average

The performance of the CWG-based waveforms was compared to both Kasami codes and a random digital signal in terms of spatial & Doppler resolution, auto- & cross-correlation sidelobes, sample rates, as well as narrowband & spoofer jamming. Kasami codes and random digital signals were chosen as the basis of comparison as they share some characteristics with the waveform from the CWG that would make them the obvious alternative operationally. First they both are spread spectrum signals that distribute the signal energy widely across their spectral support. Second, there are a large number of these codes that are approximately non-interfering, although in the Kasami case there are many less than the CWG waveform or digital noise. Finally, both types of waveforms are valued for their low auto- and cross-correlation sidelobes [7] [8].

The primary advantage to the PRA/Duke approach is that it brings some of the value of complex pulsed-Doppler waveforms, *i.e.* energy being spread across a wide frequency bandwidth, to the low-cost, easily implemented, continuous wave (CW) like radar. Other expected advantages of this approach are:

- Radar interference rejection – The orthogonal nature of the chaotic waveforms, even from identical systems, will enable multiple systems to operate in the same physical space.
- Waveform variability enables higher performance (better clutter and noise rejection) than structured signals.
- Narrowband jammer tolerance
- Spoofing jammer resistance
- Low cost – The basic waveform generator requires no processing and consists of very few parts.
- Simple implementation – No special manufacturing techniques or tuning are required to achieve the desired dynamics.
- Reliability – Dynamics are produced by a reliable process (threshold and switch) and not dependent on analog non-linearities.

2. CHAOTIC WAVEFORM GENERATOR

Our approach to a chaos radar combines concepts from Corron et al.'s [1] design with the time-delayed feedback loops to create a high-frequency piecewise-linear chaotic system. Shown in Figure 1, our device uses a linear, band-limited feedback loop and variable gain amplifier (VGA) to produce oscillations at frequencies ~1.8 GHz. Due to the time-delayed feedback, the output signal is stored for a duration of the feedback loop delay. Similar to the guard condition from [1], we include a control loop that continuously monitors the output amplitude of the feedback signal. The control loop in our system keeps the amplitude of the feedback signal from saturating the amplifier through a nonlinear switching of the feedback loop and also gives rise to a non-repeating digital-like signal.

Figure 1 show the prototype CWG. A feedback loop connects the output of a VGA (Hittite HMC287MS8) $v(t)$ with its input using a coaxial cable with time delay τ_f = 42.5 ns. The output of the VGA is also fed through a power splitter (Mini-Circuits ZFRSC-42-S) and into a control loop of delay approximately τ_c = 40 ns that uses a logarithmic amplifier (LGA) (Analog Devices AD8319) and digital TTL inverter gate (Texas Instruments SN74AUC1G04) in series. The output of the logic gate is connected to the voltage control port v_{ctl} of the VGA and is also fed into a latch circuit. The latch circuit (Texas Instruments SN74AUC1G80) introduces hysteresis to the output of the logic gate where $S(t)$ is the latch circuit state.

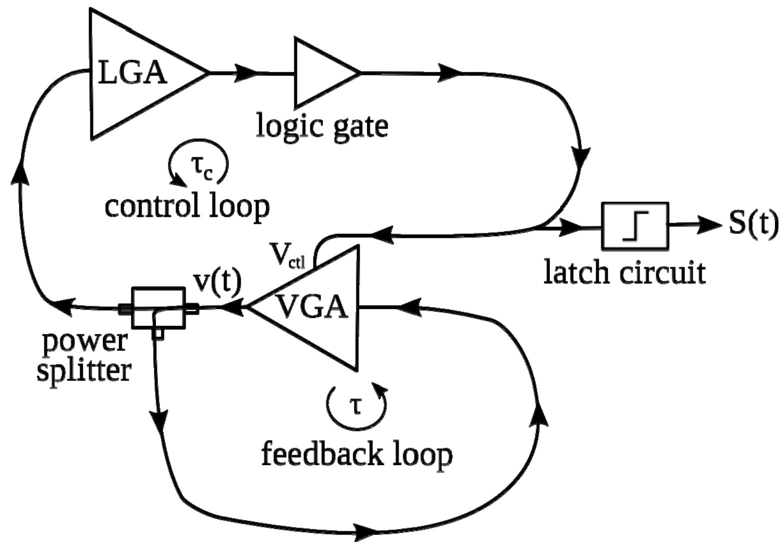

Figure 1. Prototype chaotic waveform generator. The interaction between the feedback and control loops give rise to chaotic analog and digital signals.

2.1 Experimental Results

The time evolution of the system's dynamics is monitored using an 8-GHz analog-bandwidth 40-GS/s oscilloscope (DSO80804A). In Figure 2a, a time series is plotted of the analog variable $v(t)$ that oscillates with an approximate central frequency f_c = 1.77 GHz. In Figure 2b, the power spectral density (PSD) of $v(t)$ is plotted. The bandwidth of $v(t)$ is spread symmetrically about f_c. In addition, the time evolution and the PSD of the switching state, $S(t)$, is plotted in Figure 2c and Figure 2d, respectively. From the time series in Figure 2a and Figure 2c, it can be seen that the dynamics in $v(t)$ and $S(t)$ are not periodic, and from Figure 2b and Figure 2d, it can be seen that the spectra of $v(t)$ and $S(t)$ do not overlap.

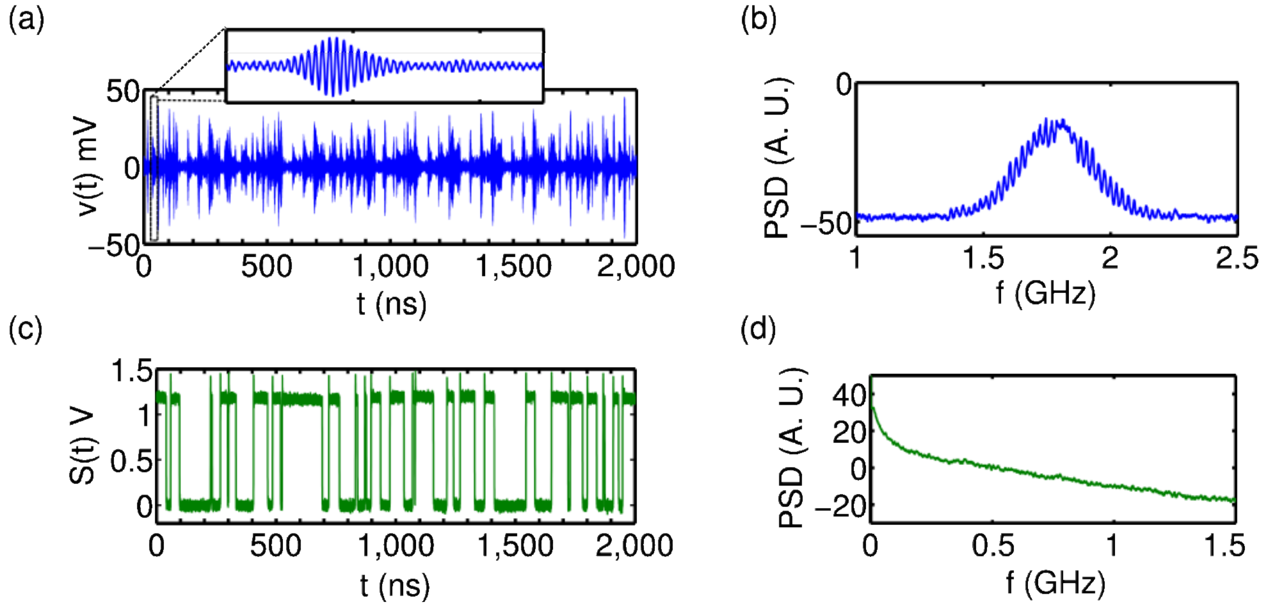

Figure 2. Outputs from the experimental system. (a) Time series and (b) PSD of $v(t)$ oscillating at high frequency with varying amplitude. A zoom of $v(t)$ is shown for t = 100-125 ns. (c) Time series and (d) PSD of switching state, $S(t)$. PSDs are plotted on a log scale and averaged over a 7.5 MHz window.

Figure 3 shows the normalized auto-correlation coefficient of $S(t)$ as a function of lag time. In the figure, the auto-correlation function has only one significant peak at lag time $t = 0$, signifying that the sequence $S(t)$ is only highly correlated with itself with zero lag. Thus, transitions in $S(t)$ do not repeat.

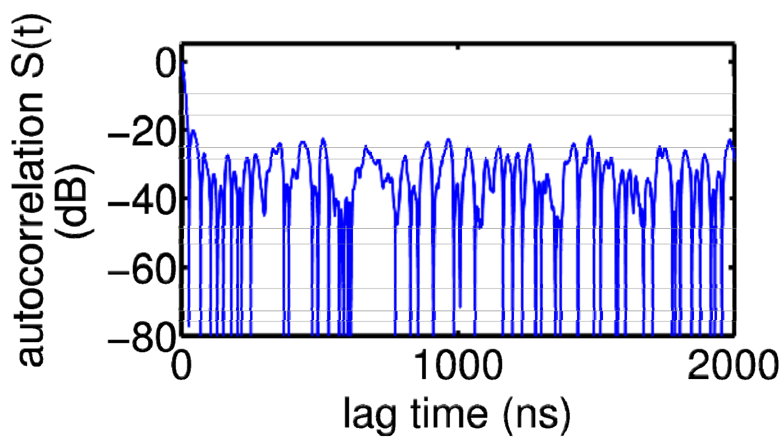

Figure 3. Auto-correlation function of the experimental switching state S(t) as a function of the time lag.

In the matched-filter concept presented by Corron *et al.* [1], the chaos and switching signals have overlapping bandwidths, and thus the transmitted radar signal contains all of the low-frequency spectral power that is associated with the digital information. In this case the matched-filter design that they developed is necessary to recover the digital information on receive. This is different from our system, where the bandpass characteristics of our feedback loop eliminate low frequencies that are associated with the switching signal, and thus the bandwidths of $v(t)$ and $S(t)$, the latch circuit state, do not overlap substantially. Therefore the digital state variable, $S(t)$, can be used directly as the CWG output without the need of the additional matched filter step.

The reduction of the signal to a phase-shifted keyed signal has some other practical advantages as it enables the use of class AB, B, and C power amplifiers, which have higher power efficiencies than the linear class A amplifiers without introducing unnecessary distortions.

3. RADAR DESIGN

Figure 4 shows a radar design based on the PRA/Duke Chaotic Waveform Generator. In this design, the CWG creates a sequence of -1's and +1's. This sequence is captured at the Nyquist rate or higher by the digital section. A simple clocked logic gate will suffice. The sequence is partitioned into pulses, up converted, transmitted, received, down converted, and then matched filtered.

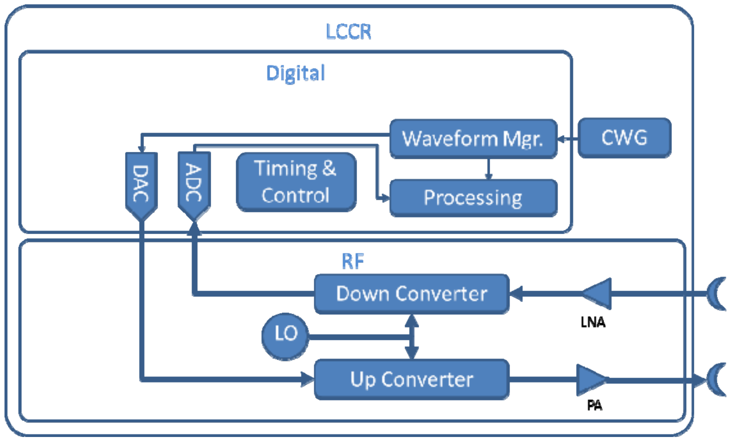

Figure 4. Radar block diagram based on the PRA/Duke CWG.

The chaos radar hardware design consists of antennas, transmitter, receiver, digital signal processor, microwave reference / L.O. frequency source, and DC battery power source. The transmitter produces 1 watt of RF power at the power amplifier output. With a commercial-off-the-shelf (COTS) low-cost, light-weight, 16-dBi gain, 25^{o} beam width dish antenna, it can easily cover a 0.5-km range with a 1-m^2 RCS target (the approximate cross-section of a person). Table 2 shows the radar range equation result.

Table 2. S- Band Chaos Radar Range Estimate – Noise Limited at 42 MHz

TX power out	1	W
TX antenna gain	16	dBi
Carrier frequency	10	GHz
Target RCS	1	m^2
RX antenna gain	16	dBi
Max. distance to target	494	m

The radar itself is a bi-phase, phase-modulated, pulsed-Doppler type radar where modulated pulses are transmitted and received and a matched filter is used to compress the pulse into a bandwidth-limited thumbtack type response. The basic waveform parameters are summarized in Table 3.

Table 3. Waveform Parameters

Parameter	**Value**	**Units**
Signal bandwidth	42 MHz	MHz
Pulse length	6	µs
Pulse in dwell	44	#
Duty cycle	100	%

Signal processing is accomplished in a digital correlation process where the received pulses are convolved with a time-reversed conjugated copy of the outgoing pulse. This efficiently accomplishes an auto-correlation in the time domain for every range bin. *N* pulses that define a dwell are aligned, and an FFT is taken in the Doppler dimension to recover a range-Doppler map. In this map, simple targets show up with individual peaks.

The performance of this radar design was explored using a combination of experimentally derived waveforms used in a Matlab simulation of the signal processing and radar environment including: noise, multiple moving targets, digitization artifacts, and non-linear amplifier effects. In addition to the experimentally derived waveform, both Kasami and random digital waveforms were implemented with bandwidths equivalent to the experimental waveform to give a basis for comparison.

3.1 Waveform Sample Rate

The digital signal from the CWG is unlike typical digital signals in which each chip has a fixed width. The temporal length of chips in this signal is continuously modulated from a minimum length up to some practical maximum. The Nyquist criterion indicates that without regard to this chip structure, sampling at twice the bandwidth of the signal is sufficient to reconstruct the waveform. However, it is not clear how this minimal sampling will affect the correlation performance.

To test this issue, the CWG waveform, which has a bandwidth of ~42 MHz but was sampled at 20 Gsps, is re-sampled at the Nyquist limit (84 Msps). An illustrative section of the temporal result of this resampling is shown in Figure 5. The original waveform sampled is shown in light gray, the re-sampled waveform is shown in black dashes. Note both waveforms are shown here with infinite-bandwidth transitions for clarity. It can be observed that the locations of the transitions between states have been moved. However, the minimum pulse length, which is the reciprocal of the bandwidth, has been unchanged while some of the longer pulses may have been lengthened or shortened.

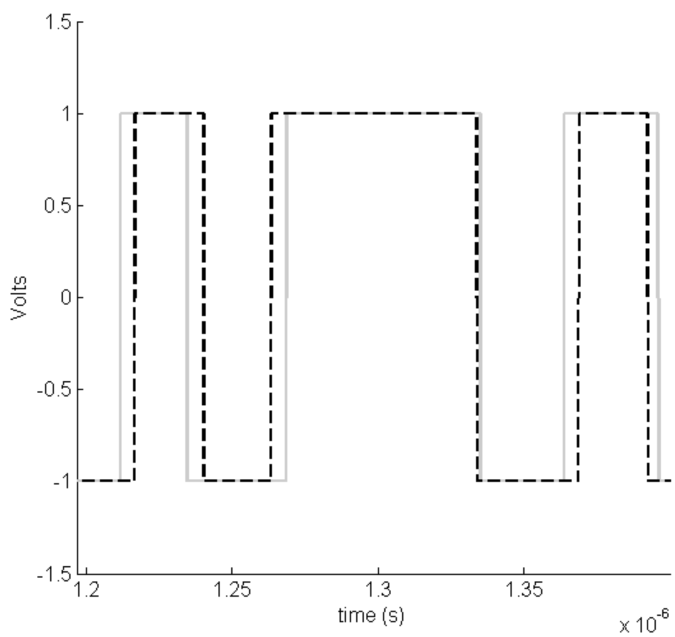

Figure 5. CWG waveform sampled at 20 Gsps (gray), re-sampled at ~84 Msps (black dashed).

Figure 6 shows the correlation of the original signal and the re-sampled signal. The black curve is the auto-correlation of the original signal, and the light gray curve is the auto-correlation of the re-sampled waveform. It can be observed that at closer ranges the two curves have identical behavior, while out in the sidelobes some differences are detectable many dB below the maximum value. The light gray curve in the figure is the cross-correlation between the original signal and the re-sampled version. Within the bandwidth of the signals, there is no functional difference between the two versions, *i.e.* they reach the same peak and have almost identical sidelobes and can be interchanged for one another.

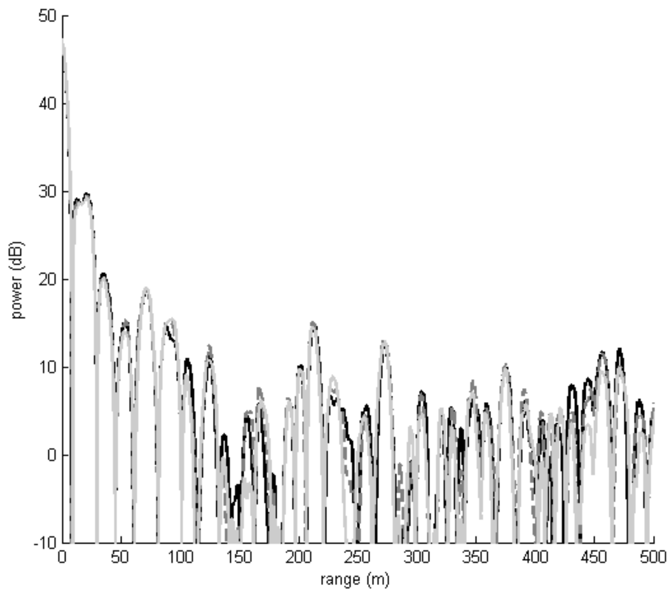

Figure 6. The correlation of the original signal with itself (black), the correlation of the original signal with the Nyquist re-sampled version (gray dashed), the correlation of the Nyquist re-sampled signal with itself (light gray).

This observation has implications for any digital system that interacts with these waveforms. Without regard for the continuous modulation of the pulse widths, these signals can be digitized without concern using the Nyquist rate.

3.2 Auto-correlation and Cross-correlation Results

A key issue in radar waveform design is the resolving power of the waveform both in ability to separate nearly spaced objects as well as the ability to not confuse the location of an object from one point in range to another. The range and Doppler resolution are important in understanding the accuracies associated with the measurement process. The basic performance of a waveform can be compared using a simple non-moving point target. The simulation was configured to compare the CWG waveform collected experimentally, a random digital waveform, and a Kasami *m*-code binary-shift coded sequence with equivalent power (0 dBm), bandwidth (42 MHz) and pulse length (~6 µs or 255 chips). The scenario that is modeled is a single target scene. The target parameters are shown in Table 4.

Table 4. Target Parameters

Parameter	Value	Units
SNR	30	dB
Target	1	#
Target range	0	m
Target velocity	0	m/s

Figure 7 illustrates the integrated performance of 44 realizations of these three waveforms. It can be seen that all the waveforms have the same width central peak, which is to be expected from equivalent bandwidths. The Kasami code auto-correlation sidelobes are 24-dB down from the peak. The CWG closest in sidelobe is approximately 18-dB down; however, the sidelobes fall off very quickly and below the Kasami levels. The random digital waveform also has sidelobes significantly below the Kasami waveform, although a secondary noise peek is located at approximately 620 m.

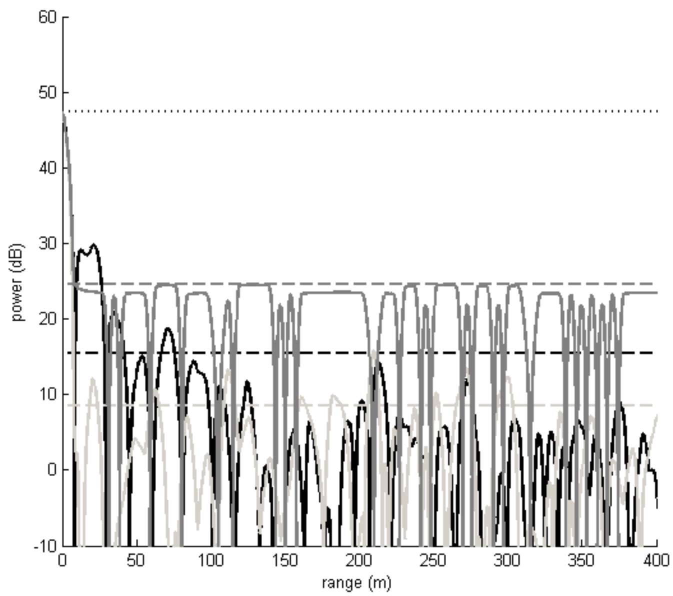

Figure 7. Range sidelobes for CWG (black), Kasami 255-length code (dark gray), and random digital waveform (light gray). The horizontal lines are the RMS values.

Recall that this analysis is based on data from the range-Doppler map where multiple pulses are averaged. For Kasami codes, every pulse is identical, and averaging does not reduce the sidelobes at all. However for both the CWG waveforms and the random digital waveforms, every pulse is different with the sidelobes varying from pulse to pulse, and averaging reduces the sidelobes by ~16 dB (or $10*\log_{10}$(number of pulses), where number of pulses is 44). Figure 8 confirms this by illustrating the auto-correlation of a single pulse for each of the waveforms. Notice that now both the random digital and CWG waveforms have higher sidelobes.

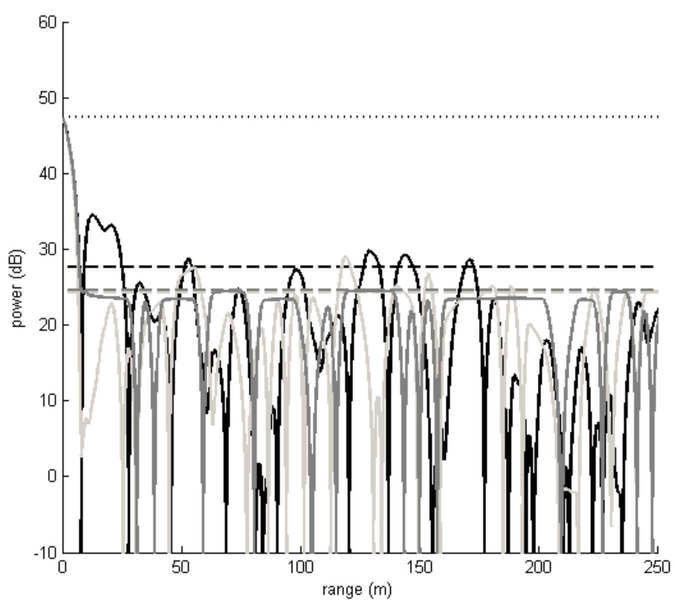

Figure 8. Single pulse auto-correlations for Kasami (dark gray), CWG (black), and random digital (light gray) waveforms as a function of sample number. The horizontal lines are the RMS values.

An important consideration in developing a class of radar waveforms is the auto-correlation of the individual waveform and cross-correlation of the individual pairs of waveforms. This is important as multiple similar radars may be required to operate in the same region. The ideal auto-correlation and cross-correlation response would be an impulse response where the values for the linear or circular convolution of waveforms would be zero except for the zero lag auto-correlation. For binary phase-coded waveforms, achieving this impulse condition is not possible. This observation is the result of a significant body of work since Welch [9] in 1974 initially developed lower bounds for binary sequences. Sarwate and Pursley [10] developed bounds that incorporate the trade between auto-correlation and cross-correlation performance.

Figure 9 shows the cross-correlation performance of the CWG, the equivalent bandwidth random digital, and Kasami waveforms. In addition, a lower bound on performance (Welch bound) for these code lengths is plotted as well. It can be seen, for a single pulse, the CWG waveform performs about 1-dB worse than both the Kasami codes and the random digital codes. However, there are only a small number of Kasami codes, approximately sqrt(code length) or 16 in this case; so the Kasami curve stops at 15 interferers. The CWG and random digital cross-correlations can continue on indefinitely. The Welch boundary curve was derived from the Welch bound for a single waveform by adding the equivalent RMS cross-correlation level for each additional interferer.

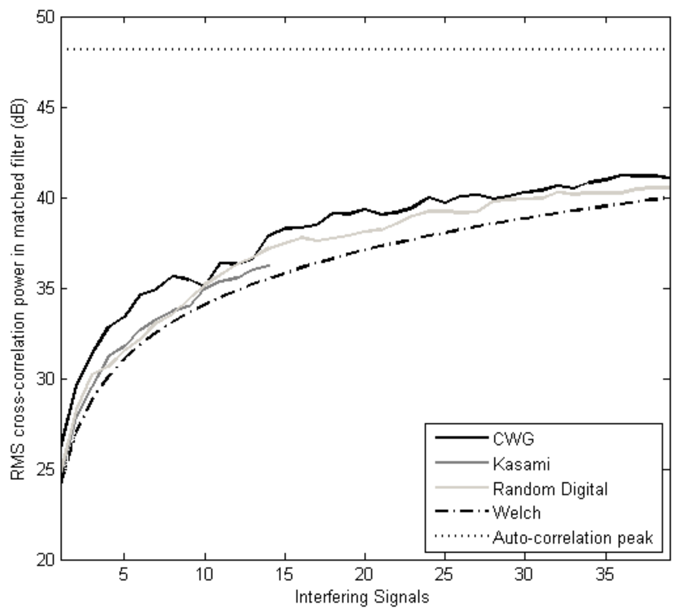

Figure 9. Cross-correlation as a function of number of interfering signals for CWG (black), Kasami-255 (dark gray), and random digital (light gray) waveforms; the Welch bound (black dash); and the peak of the auto-correlation (horizontal black dots).

This observation has a practical result. In systems using the highly structured types of waveforms, each system operating in a given theater needs to be assigned a unique code to minimize interference. This results in the need for address management systems. However, systems that use either the CWG or the random digital coding can just be turned on and a low interference level will result; no management is necessary.

3.3 Performance with Jamming

For a number of possible applications, it is necessary to understand how the radar waveforms perform under jamming conditions. There are three major classes of jamming: (1) broad-band noise jamming, which raises the noise floor, (2) narrowband jamming, and (3) smart or coherent jamming. Broad-band noise jamming attempts to directly raise the noise floor in the radar. It is typically addressed with spatial arrays and improved dynamic range. The narrowband jammer seeks to exploit the matched filter to raise the effective noise floor within the signal processor. Smart jamming uses an active receiver that acquires and records the radar signal and then replays it to mimic real targets in an effort to overwhelm any tracker processing.

3.3.1 Narrowband Jamming

To understand the relative performance of the CWG waveform in narrowband jamming conditions, the simulation was configured to generate a 10-MHz sinusoidal signal within the bandwidth of the radar signals. Figure 10 shows the effect of this jamming signal on the uncompressed spectrum of the Kasami waveform. The dark gray curve is the target response, and the dashed curve is the same response with the jamming signal added. Note that in this case, the jamming signal has the same power as the Kasami code but is more concentrated in frequency.

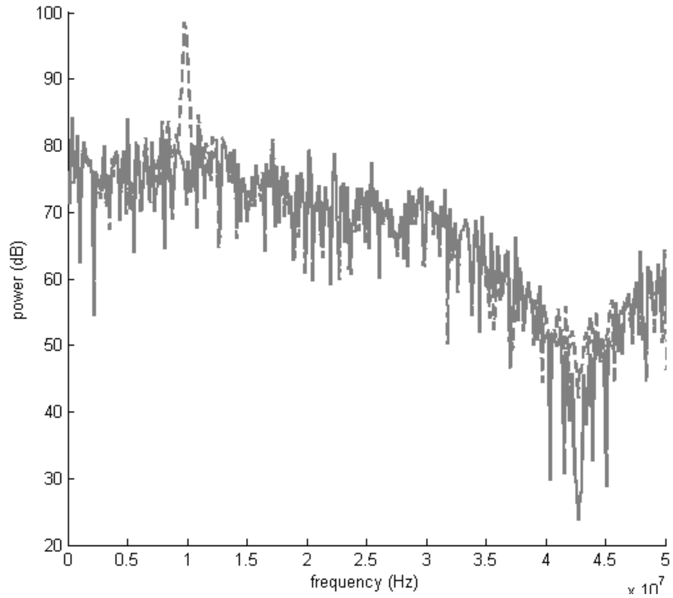

Figure 10. Illustration of narrowband jamming on a Kasami power spectrum. The dark gray curve is the Kasami code with no jamming and the dashed curve is with an equal power narrow band jammer.

Figure 11 plots the relative performance for the CWG waveform, the Kasami code, and the random digital waveform of the same bandwidth, pulse length, and power as the jammer power is swept. The performance metric chosen here is the compressed peak-to-sidelobe level. An entire 44-pulse range-Doppler map was collected, and the ratio of the peak of the true target response to the RMS range sidelobes within that Doppler bin is computed. If the sidelobe level is too high, then targets cannot be identified. Notice in all cases, the trend is similar; as the jammer power is increased, the peak–to-sidelobe level decreases. The major difference is the starting point of the three curves.

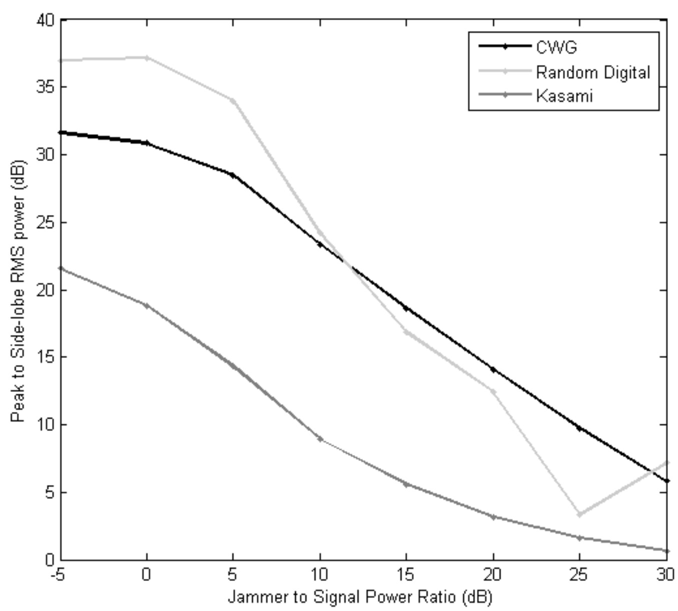

Figure 11. Narrowband jamming performance of the three waveforms.

An example of the effect of narrowband jamming on the range response of a real target for the three waveforms is shown in Figure 12. Once again, this image was produced from a full 44-pulse dwell, and the jammer-to-signal ratio was 0 dB. The figure can be compared to the "no jamming" results shown in Figure 7. From this comparison, it can be seen that the major differences are some additional fluctuations in the sidelobes. As the jammer power increases, these fluctuations increase in size and start to affect the peak as well.

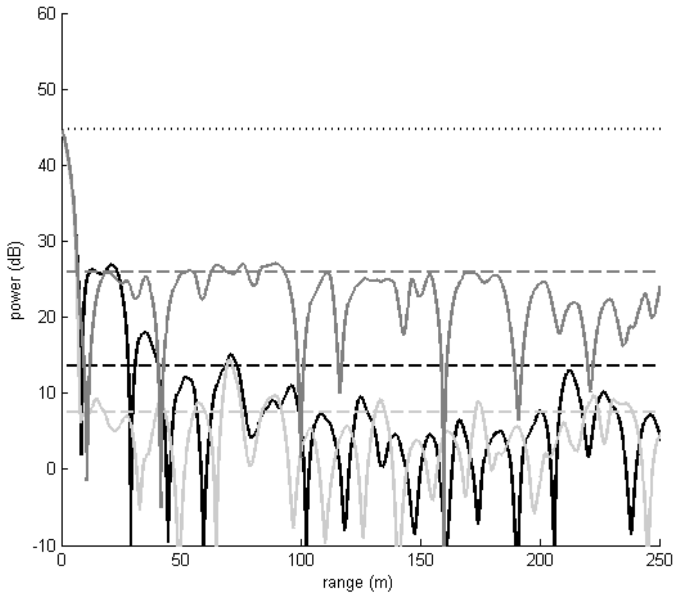

Figure 12. An example of the range sidelobes under narrowband jamming at 10 MHz of same power as the signal. This graph can be compared to the "no jamming" case in Figure 7. The CWG-waveform is black, the Kasami-255 is dark gray, and the random digital is light gray. The horizontal lines are the RMS values.

3.3.2 Coherent Jammers

Smart jamming or coherent Electronic Attack (EA) techniques rely on deterministic concepts that depend upon the nature of transmitted waveforms. Once these waveforms have been detected, it is relatively straightforward for a coherent digital RF memory (DRFM) jammer to emulate by playing back a copy of the waveform modified to look like a target response and confuse the ground-based radar. A DRFM jammer can be capable of generating multiple false target tracks in the radar track processor forcing the radar to dedicate precious resources to false targets, and thus endangering soldiers and assets by degrading detection and track performance of actual targets. This type of jamming is an increasing concern as the tools to implement it are available online and require only a little electrical engineering expertise.

One way this jamming process is accomplished is by the jammer recording the first pulse in a dwell and then determining the timing between pulses. With this information, the jammer can emulate a target at any range and with any Doppler offset, as long as the radar waveform processing does not "notice" the lack of a first pulse return.

For both the CWG and the random digital waveforms, the order and timing of the pulses is critical for the successful compression of the pulse. So shifting the pulses in a dwell by one pulse prevents compression. This is not the case with a typical Kasami type system, where every pulse is identical.

Figure 13 shows the effect of shifting the pulse in a dwell by one for all the waveforms under consideration. In this case, the simulation was configured to generate a target return at 200 m with a velocity of 570 m/s. An example jammer was configured to generate a false target at 0 m with no Doppler after a single pulse shift. The true target response was held constant and the jammer power was swept. The vertical axis shows the ratio of the true target power to the false target power at the peak of their responses. It can be seen that the compression of the Kasami waveform could be easily fooled. However, both the CWG and the random digital waveforms suppressed the jammer by many dB.

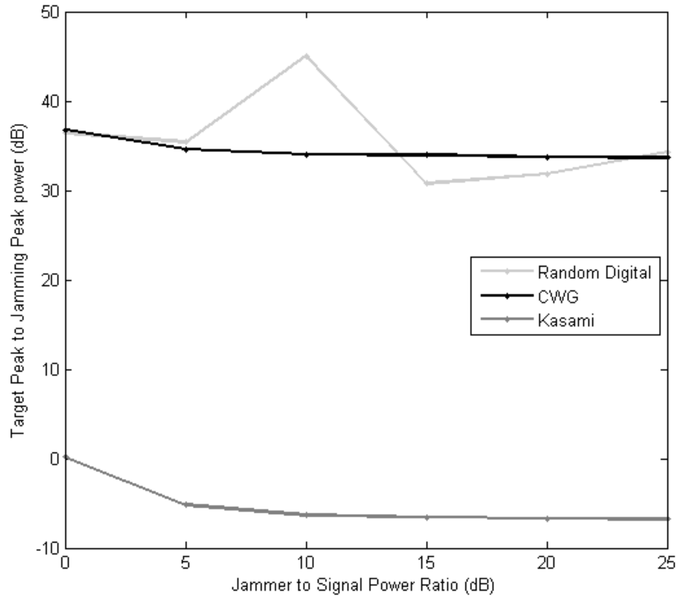

Figure 13. Coherent jammer performance for Kasami (dark gray), random digital (light gray) and CWG-waveforms (black).

3.4 Waveform Variability

The random phase waveform is simple to create with any random noise generator and generates reasonably low-correlation sidelobes. In general, the correlation sidelobe behavior for the PRA/Duke chaotic waveform is similar to that of the random phase waveform. However, there is one distinction between the two waveforms. For random waveforms that use a code length of N bits, the total number of available waveforms is 2^N of which only a subset provide sufficient randomness for adequate correlation performance. For low bandwidth radars, the code lengths are typically short and thus the number of random waveforms will be limited. For Kasami codes of length 2^N-1 (with N an even integer), there are $2^{N/2}$-1 total codes. However, due to the increased number of degrees of randomness in a chaotic code, there are significantly more codes for a given length than random phase or Kasami codes. In addition, for short code lengths, the performance of random codes will depend upon how random the code is. In other words, sometimes a random process can generate phase states that are highly correlated and that will not occur for the chaotic codes. Having a large set of codes that generally have good and similar correlation behavior is a significant benefit of chaotic codes for some radar applications.

The potential advantage of the CWG can be seen in the number of unique codes available to each waveform type as shown in Table 5. This table assumes a Nyquist sampling rate where the smallest chip width consists of 2 samples. For example, a Kasami 255-chip code has 510 samples and should be compared with digital random codes of 510 samples (255 length) and CWG waveforms with 510 samples. It can be seen that the total number of unique codes for the CWG case is much larger that the corresponding number of Kasami or random codes. This can be traced to the observation that a CWG chip can be an odd number of samples, whereas the other chips are constrained to be an even number of samples. This extra variability results in a very large set of possible codes, not all with good mutual correlation properties. However, the attractor structure of the CWG results in a reduced set of used codes. Therefore the goal of the CWG design is to have the attractor cover the set of waveforms with good auto- and cross-correlation properties.

Table 5. Number of unique codes in each waveform for a fixed code length sampled at a Nyquist sampling rate.

Clocked Code Length	# of possible codes		
	CWG	Random Digital	Kasami
2	4	4	-
3	10	8	-
4	26	16	2
…	…	…	-
10	8362	1024	-
…	…	…	-
16	$\sim 10^6$	65,536	4
…	…	…	-

4. CONCLUSIONS

Our proposed radar design is based on a time-delay feedback chaotic waveform generator that is easy to create and provides waveforms with many attractive features including: the ability to operate multiple radars in the same area, good range performance in a pulsed-Doppler design, jammer tolerance, and simple implementation. The near-in range sidelobe performance of the CWG waveform could be improved. In addition, this effort did not address the computational cost of implementing matched filtering for continuously evolving pulses within a dwell. In a planned future effort, PRA and Duke will refine these designs, build a working prototype radar, and verify its performance in a field environment.

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